

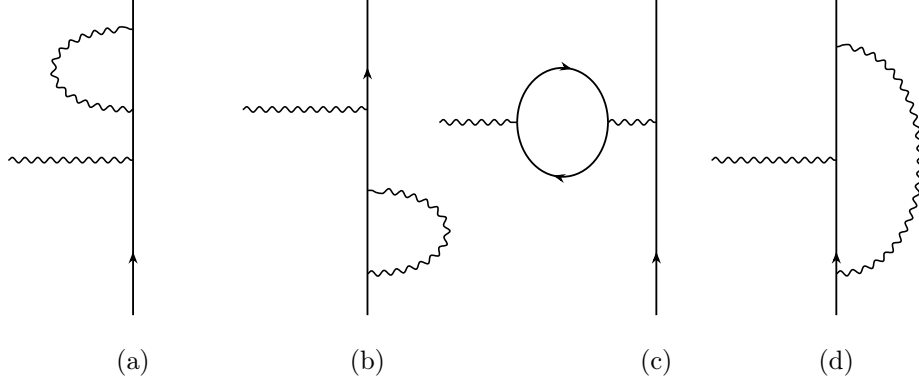


Figure 11.4. One-loop diagrams for the photon-lepton vertex function $\Gamma^{\mu*}$. Here wavy lines represent photons; other lines represent electrons or muons. Diagrams (a) and (b) are cancelled by lepton field renormalization terms; diagram (c) arises from the vacuum polarization calculated in Section 11.2; and (d) is the term calculated in Section 11.3.

11.3 Anomalous Magnetic Moments and Charge Radii

For our next example, we shall calculate the shift in the magnetic moment and the charge radius of an electron or muon due to lowest-order radiative corrections. The one-loop graphs and renormalization corrections for the photon-lepton vertex are shown in Figure 11.4. Of these graphs, those involving insertions in incoming or outgoing lepton lines vanish because the lepton is on the mass shell, as discussed in Section 10.3. The graph involving an insertion in the external photon line is the vacuum polarization effect, discussed in the previous section. This leaves one one-loop graph (the last in Figure 11.4) that needs to be calculated here:

$$\Gamma_{1\text{ loop}}^{\mu}(p', p) = \int d^4k [e\gamma^{\rho}(2\pi)^4] \left[\frac{-i}{(2\pi)^4} \frac{-i(\not{p}' - \not{k}) + m}{(p' - k)^2 + m^2 - i\epsilon} \right] [\gamma^{\mu}] \\ \times \left[\frac{-i}{(2\pi)^4} \frac{-i(\not{p} - \not{k}) + m}{(p - k)^2 + m^2 - i\epsilon} \right] [e\gamma_{\rho}(2\pi)^4] \left[\frac{-i}{(2\pi)^4} \frac{1}{k^2 - i\epsilon} \right], \quad (11.3.1)$$

where p' and p are the final and initial lepton four-momenta, respectively. (The contribution of the vertex connecting the external photon line and the internal lepton line is taken as γ^{μ} , because a factor $e(2\pi)^4$ was extracted in defining Γ^{μ} .)

This integral has an obvious ultraviolet divergence, roughly like $\int d^4k/(k^2)^2$. Unlike the case of the vacuum polarization, here we do not need a fancy regularization procedure like dimensional regularization to maintain the structure required by gauge invariance, because the photon is a neutral particle and so the integral may be rendered finite by suitable modifications of the photon propagator (for instance by including a factor $M^2/(k^2 + M^2)$ with a large cutoff mass M), without having to introduce modifications elsewhere to maintain gauge invariance. In any case, as we shall see, the anomalous magnetic moment and charge radii can be calculated without encountering any ultraviolet divergences at all. In what follows we shall leave the integrals for the vertex function in their infinite form, with it being understood that if necessary any divergent integrals can be expressed in terms of a cutoff mass M .

We start by combining denominators, using a repeated version of the Feynman trick described in the Appendix to this chapter

$$\frac{1}{ABC} = 2 \int_0^1 dx \int_0^x dy [Ay + B(x-y) + C(1-x)]^{-3}. \quad (11.3.2)$$

Applied to the denominators in Eq. (11.3.1), this gives

$$\begin{aligned} & \frac{1}{(p' - k)^2 + m^2 - i\epsilon} \frac{1}{(p - k)^2 + m^2 - i\epsilon} \frac{1}{k^2 - i\epsilon} \\ &= 2 \int_0^1 dx \int_0^x dy \left[((p' - k)^2 + m^2 - i\epsilon)y + ((p - k)^2 + m^2 - i\epsilon)(x - y) \right. \\ & \quad \left. + (k^2 - i\epsilon)(1 - x) \right]^{-3} \\ &= 2 \int_0^1 dx \int_0^x dy \left[(k - p'y - p(x - y))^2 + m^2 x^2 + q^2 y(x - y) - i\epsilon \right]^{-3}, \end{aligned} \quad (11.3.3)$$

where $q \equiv p - p'$ is the momentum transferred to the photon. Shifting the variable of integration

$$k \longrightarrow k + p'y + p(x - y)$$

the integral (11.3.1) becomes

$$\begin{aligned} \Gamma_{1 \text{ loop}}^\mu(p', p) &= \frac{2ie^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int \frac{d^4 k}{[k^2 + m^2 x^2 + q^2 y(x - y) - i\epsilon]^3} \\ & \quad \times \gamma^\rho [-i(\not{p}'(1 - y) - \not{k} - \not{p}(x - y)) + m] \gamma^\mu \\ & \quad \times [-i(\not{p}(1 - x + y) - \not{k} - \not{p}'y) + m] \gamma_\rho. \end{aligned} \quad (11.3.4)$$

Our next step is a Wick rotation. As explained in the previous section, the $-i\epsilon$ in the denominator dictates that when we rotate the k^0 contour of integration to the imaginary axis we must rotate counterclockwise, so that the integral over k^0 from $-\infty$ to $+\infty$ is replaced with an integral over imaginary values from $-i\infty$ to $+i\infty$, or equivalently over real values of $k^4 \equiv -ik^0$ from $-\infty$ to $+\infty$. We also exploit the rotational symmetry of the denominator in Eq. (11.3.4); we drop terms in the denominator of odd order in k , replace $k^\mu k^\nu$ with $\eta^{\mu\nu} \kappa^2/4$, and replace the volume element $d^4 k = i dk^1 dk^2 dk^3 dk^4$ with $2i\pi^2 \kappa^3 d\kappa$, where κ is the Euclidean length of the four-vector k . Putting this all together, Eq. (11.3.4) now becomes

$$\begin{aligned} \Gamma_{1 \text{ loop}}^\mu(p', p) &= \frac{-4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \left\{ -\frac{\kappa^2}{4} \gamma^\rho \gamma^\sigma \gamma^\mu \gamma_\sigma \gamma_\rho \right. \\ & \quad \left. + \gamma^\rho [-i(\not{p}'(1 - y) - \not{p}(x - y)) + m] \gamma^\mu [-i(\not{p}(1 - x + y) - \not{p}'y) + m] \gamma_\rho \right\} \\ & \quad \times [\kappa^2 + m^2 x^2 + q^2 y(x - y)]^{-3}. \end{aligned} \quad (11.3.5)$$

We are interested here only in the matrix element $\bar{u}' \Gamma^\mu u$ of the vertex function between Dirac spinors that satisfy the relations

$$\bar{u}' [i \not{p}' + m] = 0, \quad [i \not{p} + m] u = 0.$$

We can therefore simplify this expression by using the anticommutation relations of the Dirac matrices to move all factors $\not{p}'$ to the right and all factors $\not{p}$ to the left, replacing them when they arrive on the right or left with im . After a straightforward but tedious calculation, Eq. (11.3.5) then becomes

$$\begin{aligned} \bar{u}' \Gamma_{\text{one loop}}^\mu(p', p) u = & \frac{-4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \bar{u}' \left\{ \gamma^\mu [-\kappa^2 + 2m^2(x^2 - 4x + 2) \right. \\ & + 2q^2(y(x - y) + 1 - x)] \\ & + 4imp'^\mu(y - x + xy) + 4imp^\mu(x^2 - xy - y) \left. \right\} u \\ & \times [\kappa^2 + m^2x^2 + q^2y(x - y)]^{-3}. \end{aligned} \quad (11.3.6)$$

We next exploit the symmetry of the final factor under the reflection $y \rightarrow x - y$. Under this reflection, the functions $y - x + xy$ and $x^2 - xy - y$ that multiply p'^μ and p^μ are interchanged, so both may be replaced with their average:

$$\frac{1}{2}(y - x + xy) + \frac{1}{2}(x^2 - xy - y) = -\frac{1}{2}x(1 - x).$$

This gives finally

$$\begin{aligned} \bar{u}' \Gamma_{\text{one loop}}^\mu(p', p) u = & \frac{-4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \\ & \times \bar{u}' \left\{ \gamma^\mu [-\kappa^2 + 2m^2(x^2 - 4x + 2) + 2q^2(y(x - y) + 1 - x)] \right. \\ & \left. - 2im(p'^\mu + p^\mu)x(1 - x) \right\} u [\kappa^2 + m^2x^2 + q^2y(x - y)]^{-3}. \end{aligned} \quad (11.3.7)$$

Note that p^μ and p'^μ now enter only in the combination $p^\mu + p'^\mu$, as required by current conservation.

There are other diagrams that need to be taken into account. Of course, there is the zeroth-order term γ^μ in Γ^μ . The term proportional to $Z_2 - 1$ in the correction term (11.1.9) yields a term in Γ^μ

$$\Gamma_{\mathcal{L}_2}^\mu = (Z_2 - 1)\gamma^\mu. \quad (11.3.8)$$

Also, the effect of insertions of corrections to the external photon propagator is a term:

$$\Gamma_{\text{vac pol}}^\mu(p', p) = \frac{1}{(p' - p)^2 - i\epsilon} \Pi^{\mu\nu}(p' - p) \gamma_\nu. \quad (11.3.9)$$

The form of each of these terms is in agreement with the general result (10.6.10) (with $H(q^2) = 0$)

$$\bar{u}' \Gamma^\mu(p', p) u = \bar{u}' \left[\gamma^\mu F(q^2) - \frac{i}{2m} (p + p')^\mu G(q^2) \right] u. \quad (11.3.10)$$

To order e^2 , the form factors are

$$\begin{aligned} F(q^2) = & Z_2 + \pi(q^2) + \frac{4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \\ & \times \frac{\kappa^2 - 2m^2(x^2 - 4x + 2) - 2q^2(y(x - y) + 1 - x)}{[\kappa^2 + m^2x^2 + q^2y(x - y)]^3}, \end{aligned} \quad (11.3.11)$$

$$G(q^2) = \frac{-4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \frac{4m^2 x(1-x) \kappa^3 d\kappa}{[\kappa^2 + m^2 x^2 + q^2 y(x-y)]^3}, \quad (11.3.12)$$

where $\pi(q^2)$ is the vacuum polarization function (11.2.22).

The integral for the form factor $G(q^2)$ is finite as it stands:

$$G(q^2) = -\frac{e^2 m^2}{4\pi^2} \int_0^1 dx \int_0^x dy \frac{x(1-x)}{m^2 x^2 + q^2 y(x-y)}. \quad (11.3.13)$$

This makes it easy to calculate the anomalous magnetic moment. As noted in Section 10.6, it is only the γ^μ term that contributes to the magnetic moment, so the effect of radiative corrections is to multiply the Dirac value $e/2m$ of the magnetic moment by a factor $F(0)$. But the definition of e as the true lepton charge requires that

$$F(0) + G(0) = 1, \quad (11.3.14)$$

so the magnetic moment may be expressed as

$$\mu = \frac{e}{2m} (1 - G(0)). \quad (11.3.15)$$

From Eq. (11.3.13), we find

$$-G(0) = \frac{e^2}{8\pi^2} = 0.001161. \quad (11.3.16)$$

This is the famous $\alpha/2\pi$ correction first calculated by Schwinger [5].

Of course, this is only the first term in the radiative corrections to the magnetic moment. Even in just the next order, fourth order in e , there are so many terms that the calculations become quite complicated. However, because of the large muon–electron mass ratio, there is one fourth-order term in the magnetic moment of the muon that is somewhat larger than any of the others. It arises from the insertion of an electron loop in the virtual photon line of the second-order diagram, as shown in Figure 11.5. The effect of this electron loop is

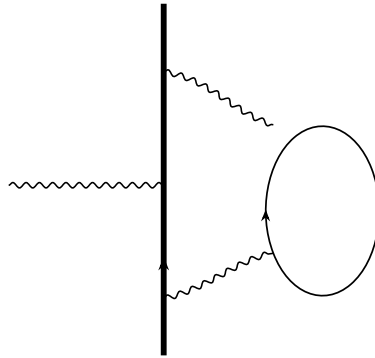


Figure 11.5. A two-loop diagram for the muon magnetic moment. Here the heavy straight line represents a muon; the light wavy lines are photons; and the other light lines are electrons. This diagram makes a relatively large contribution to the fourth-order muon gyromagnetic ratio, proportional to $\ln(m_\mu/m_e)$.

to change the photon propagator $1/k^2$ in Eq. (11.3.1) to $(1 + \pi_e(k^2))/k^2$, where $\pi_e(k^2)$ is given by Eq. (11.2.22), but with the mass m taken as the electron mass:

$$\pi_e(k^2) = \frac{e^2}{2\pi^2} \int_0^1 x(1-x) \ln \left(1 + \frac{k^2 x(1-x)}{m_e^2} \right) dx.$$

Inspection of Eq. (11.3.12) shows that in calculating the muon magnetic moment the effective cutoff on the virtual photon momentum k is m_μ . The ratio m_μ/m_e is so large that for k^2 of order m_μ^2 we may approximate

$$\pi_e(k^2) \simeq \frac{e^2}{2\pi^2} \int_0^1 x(1-x) \ln(m_\mu^2/m_e^2) = \frac{e^2}{12\pi^2} \ln(m_\mu^2/m_e^2), \quad (11.3.17)$$

with the neglected terms having coefficients of order unity in place of $\ln(m_\mu^2/m_e^2)$. Since this is a constant, the change in $-G(0)$ produced by adding an electron loop in the virtual photon line is simply given by multiplying our previous result (11.3.16) for $-G(0)$ by Eq. (11.3.17), so that now

$$\mu_\mu = \frac{e}{2m_\mu} \left(\frac{e^2}{8\pi^2} + \frac{e^4}{96\pi^4} \left[\ln \frac{m_\mu^2}{m_e^2} + O(1) \right] \right). \quad (11.3.18)$$

(As we shall see in Volume II, this argument is a primitive version of the method of the renormalization group.) The result (11.3.18) may be compared with the full fourth-order result [6]

$$\mu_\mu = \frac{e}{2m_\mu} \left\{ \frac{e^2}{8\pi^2} + \frac{e^4}{96\pi^4} \left[\ln \frac{m_\mu^2}{m_e^2} - \frac{25}{6} + \frac{197}{24} + \frac{\pi^2}{2} + \frac{9\zeta(3)}{2} - 3\pi^2 \ln 2 + O\left(\left(\frac{m_e}{m_\mu}\right)^2\right) \right] \right\}. \quad (11.3.19)$$

It turns out that the $O(1)$ terms multiplying $e^4/96\pi^4$ add up to -6.137 , which is not very much smaller than $\ln(m_\mu^2/m_e^2) = 10.663$, so the approximation (11.3.18) gives the fourth-order terms only to a factor of order 2. The correct fourth-order result (11.3.19) gives $\mu_\mu = 1.00116546e/2m_\mu$, in comparison with the second-order result $\mu_\mu = 1.001161e/2m_\mu$ and the current experimental value [7], $\mu_\mu = 1.001165923(8)e/2m_\mu$.

Now let us turn to the other form factor. The integral in Eq. (11.3.11) for $F(q^2)$ has an ultraviolet divergence. However, in order to satisfy the charge-non-renormalization condition (11.3.14), it is necessary that Z_2 take the value

$$Z_2 = 1 + \frac{e^2}{8\pi^2} - \frac{4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \frac{\kappa^2 - 2m^2(x^2 - 4x + 2)}{[\kappa^2 + m^2 x^2]^3}. \quad (11.3.20)$$

(Recall that $\pi(0) = 0$.) This is itself ultraviolet divergent, with an infinite part

$$(Z_2 - 1)_\infty = -\frac{e^2}{8\pi^2} \int^\infty \frac{d\kappa}{\kappa}. \quad (11.3.21)$$

Inserting Eq. (11.3.20) back into Eq. (11.3.11) gives

$$F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{4\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \int_0^\infty \kappa^3 d\kappa \\ \times \left\{ \frac{\kappa^2 - 2m^2(x^2 - 4x + 2) - 2q^2[y(x - y) + 1 - x]}{[\kappa^2 + m^2x^2 + q^2y(x - y)]^3} - \frac{\kappa^2 - 2m^2(x^2 - 4x + 2)}{[\kappa^2 + m^2x^2]^3} \right\}. \quad (11.3.22)$$

The integral over κ is now convergent:

$$F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \left\{ \frac{-m^2[x^2 - 4x + 2] - q^2[y(x - y) + 1 - x]}{m^2x^2 + q^2y(x - y)} \right. \\ \left. + \frac{x^2 - 4x + 2}{x^2} - \ln \left[\frac{m^2x^2 + q^2y(x - y)}{m^2x^2} \right] \right\}. \quad (11.3.23)$$

However, we see that the integral over x and y now diverges logarithmically at $x = 0$ and $y = 0$, because there are two powers of x and/or y in the denominators, and just two differentials $dx dy$ in the numerator. This divergence can be traced to the vanishing of the denominator $[\kappa^2 + m^2x^2 + q^2y(x - y)]^3$ in Eq. (11.3.11) at $x = 0$, $y = 0$, and $\kappa = 0$. Because this infinity comes from the region of small rather than large κ , it is termed an *infrared divergence* rather than an ultraviolet divergence.

We shall give a comprehensive treatment of the infrared divergences in Chapter 13. It will be shown there that infrared divergences in the cross-section for processes like electron-electron scattering, such as those that are introduced by the infrared divergence in the electron form factor $F(q^2)$, are cancelled when we include the emission of low-energy photons as well as elastic scattering. Also, as we shall see in Chapter 14, when we calculate radiative corrections to atomic energy levels the infrared divergence in $F(q^2)$ is cut off because the bound electron is not exactly on the free-particle mass shell. For the present we shall continue our calculation by simply introducing a fictitious photon mass μ to cut off the infrared divergence in $F(q^2)$, leaving it for Chapter 14 to see how to use this result.

With a photon mass μ , the denominator $k^2 - i\epsilon$ in Eq. (11.3.1) would be replaced with $k^2 + \mu^2 - i\epsilon$. The effect would then be to add a term $\mu^2(1 - x)$ to the cubed quantity in the denominators of Eqs. (11.3.3)–(11.3.7), (11.3.11), (11.3.20), and (11.3.22). Eq. (11.3.23) then is replaced with

$$F(q^2) = 1 + \frac{e^2}{8\pi^2} + \pi(q^2) + \frac{2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \left\{ \frac{-m^2[x^2 - 4x + 2] - q^2[y(x - y) + 1 - x]}{m^2x^2 + q^2y(x - y) + \mu^2(1 - x)} \right. \\ \left. + \frac{m^2[x^2 - 4x + 2]}{m^2x^2 + \mu^2(1 - x)} - \ln \left[\frac{m^2x^2 + q^2y(x - y) + \mu^2(1 - x)}{m^2x^2 + \mu^2(1 - x)} \right] \right\}. \quad (11.3.24)$$

This integral is now completely convergent. It can be expressed in terms of Spence functions, but the result is not particularly illuminating. For our purposes in Chapter 14, it will be

sufficient to calculate the behavior of $F(q^2)$ for small q^2 . We already know from the Ward identity that $F(0) = 1 - G(0) = 1 + e^2/8\pi^2$, so let us consider the first derivative $F'(q^2)$ at $q^2 = 0$. According to Eq. (11.3.24), this is

$$F'(0) = \pi'(0) + \frac{2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \int_0^x dy \left\{ \frac{2y(x-y) + 1-x}{m^2 x^2 + \mu^2(1-x)} + \frac{m^2[x^2 - 4x + 2]y(x-y)}{[m^2 x^2 + \mu^2(1-x)]^2} \right\}. \quad (11.3.25)$$

The vacuum polarization contribution is given by Eq. (11.2.22) as

$$\pi'(0) = \frac{e^2}{60\pi^2 m^2}. \quad (11.3.26)$$

Dropping all terms proportional to powers of μ/m in Eq. (11.3.25), we then have¹

$$F'(0) = \frac{e^2}{24\pi^2 m^2} \left[\ln \left(\frac{\mu^2}{m^2} \right) + \frac{2}{5} + \frac{1}{4} \right] \quad (11.3.27)$$

with the term $2/5$ the contribution of vacuum polarization. On the other hand, Eq. (11.3.13) shows that $G(q^2)$ has a finite derivative at $q^2 = 0$,

$$G'(0) = \frac{e^2}{48\pi^2 m^2}. \quad (11.3.28)$$

These results are most conveniently expressed in terms of the charge form factor $F_1(q^2)$, defined by the alternative representation (10.6.15) of the vertex function

$$\begin{aligned} & \bar{u}(p', \sigma') \Gamma^\mu(p', p) u(p, \sigma) \\ &= \bar{u}(p', \sigma') \left[\gamma^\mu F_1(q^2) + \frac{1}{2} i [\gamma^\mu, \gamma^\nu] (p' - p)_\nu F_2(q^2) \right] u(p, \sigma). \end{aligned} \quad (11.3.29)$$

According to Eqs. (10.6.17) and (10.6.18),

$$F_1(q^2) = F(q^2) + G(q^2). \quad (11.3.30)$$

For $|q^2| \ll m^2$, this form factor is approximately

$$F_1(q^2) \simeq 1 + \frac{e^2}{24\pi^2} \frac{q^2}{m^2} \left[\ln \left(\frac{\mu^2}{m^2} \right) + \frac{2}{5} + \frac{3}{4} \right]. \quad (11.3.31)$$

This may be expressed in terms of a *charge radius* a , defined by the limiting behavior of the charge form factor for $q^2 \rightarrow 0$:

$$F_1(q^2) \rightarrow 1 - q^2 a^2 / 6. \quad (11.3.32)$$

¹The y -integral is trivial. The x -integral is most easily calculated in the limit $\mu \ll m$ by dividing the range of integration into two parts, one from 0 to s , where $\mu/m \ll s \ll 1$, and the second from s to 1.

(This definition is motivated by the fact that the average of $\exp(i\mathbf{q} \cdot \mathbf{x})$ over a spherical shell of radius a goes as $1 - \mathbf{q}^2 a^2/6$ for $\mathbf{q}^2 a^2 \ll 1$.) We see that the charge radius of the electron is given by

$$a^2 = -\frac{e^2}{4\pi^2 m^2} \left[\ln \left(\frac{\mu^2}{m^2} \right) + \frac{2}{5} + \frac{3}{4} \right]. \quad (11.3.33)$$

We will see in Chapter 14 that for electrons in atoms the role of the photon mass is played by an effective infrared cutoff that is much less than m , so the logarithm here is large and negative, yielding a positive value for a^2 .